

Nonoperating Items, Provisions, and Reserves

To project free cash flow, you would typically focus on operating expenses, such as cost of sales, distribution expenses, selling expenses, and administrative expenses. But what about nonoperating expenses, such as business realignment expenses, goodwill impairment, and extraordinary items?

Nonoperating expenses are infrequent or unusual charges that are indirectly related to the company's typical activities and not expected to recur. The conventional wisdom is that discounted-cash-flow (DCF) calculations should ignore nonoperating expenses as backward-looking, one-time costs. Yet research shows that the type and accounting treatment of nonoperating expenses can affect future cash flow and in certain situations must be incorporated into your valuation.

This chapter analyzes the most common nonoperating expenses. These include the amortization of acquired intangibles, restructuring charges, unusual charges such as litigation expenses, asset write-offs, and goodwill impairments. Since noncash expenses will be accompanied by a corresponding provision, we create a classification system of various provisions and describe the process for reorganizing the income statement and balance sheet to reflect the true effect of such provisions, if any, on company value. We show how to treat provisions in free cash flow and equity valuation.

NONOPERATING EXPENSES AND ONE-TIME CHARGES

Given their infrequent nature, nonoperating expenses and one-time charges can distort a company's historical financial performance and consequently distort our view of the future. It is therefore critical to separate one-time

nonoperating expenses from ongoing operating expenses. The idea sounds simple, but implementing it can be tricky. Nonoperating expenses are often spread across the income statement, and some are hidden within other accounts and can be discovered only by searching the company's notes. Even after you've properly identified nonoperating expenses, the job is not done. Each nonoperating expense must be carefully analyzed to determine its impact on future cash flow, and if necessary, forecasts must be adjusted to reflect any information embedded in the expense.

To assess the impact of nonoperating expenses and incorporate their information in cash flow forecasts, we recommend a three-step process:

1. *Separate operating from nonoperating items.* This process requires judgment. As a general rule, treat items that grow in line with revenues and are related to running the core business as operating. For line items that are lumpy but only tangentially related to core operations, test the impact of each line item on long-term ROIC.
2. *Search the notes for embedded one-time items.* Not every one-time charge will be separately disclosed on the income statement. Sometimes the management discussion and analysis section of the annual report will disclose additional information on one-time items.
3. *Analyze each nonoperating item for its impact on future operations.* Line items not included in earnings before interest, taxes, and amortization (EBITA) will not be included in free cash flow (FCF), so they are not part of core operating value. Therefore, it is critical to analyze each nonoperating line item separately and determine whether the charge is likely to continue in the future, in which case it should be incorporated into FCF projections.

Separating Operating from Nonoperating Expenses

Many companies include a line item on their income statement that reads "Operating income (loss)" or "Operating profit/loss." For example, in Exhibit 21.1, the income statement for Boston Scientific shows that in 2018 the company reported an operating profit of \$1.5 billion. But is this profit an accurate reflection of the company's long-run earnings potential? The accounting definition of operating profit differs from our definition of EBITA, in that the accounting standards for classifying items as nonoperating (i.e., to be recorded below operating profit or loss) are extremely strict. To benchmark core operations effectively, EBITA and net operating profit after taxes (NOPAT) should include only items related to the ongoing core business, regardless of their classification by accounting standards.

EXHIBIT 21.1 **Boston Scientific: Income Statement**

\$ million							
Accounting income statement ¹	2016	2017	2018	Reorganized income statement	2016	2017	2018
Net sales	8,386	9,048	9,823	Net sales	8,386	9,048	9,823
Cost of products sold	(2,424)	(2,593)	(2,813)	Cost of products sold	(2,424)	(2,593)	(2,813)
Gross profit	5,962	6,455	7,010	Gross profit	5,962	6,455	7,010
SG&A expense	(3,099)	(3,294)	(3,569)	SG&A expense	(3,099)	(3,294)	(3,569)
R&D expense	(920)	(997)	(1,113)	R&D expense	(920)	(997)	(1,113)
Royalty expense	(79)	(68)	(70)	Royalty expense	(79)	(68)	(70)
Amortization expense	(545)	(565)	(599)	EBITA	1,864	2,096	2,258
Intangible-asset impairment charges	(11)	(4)	(35)				
Contingent consideration benefit	(29)	80	21				
Restructuring charges	(28)	(37)	(36)				
Litigation-related charges	(804)	(285)	(103)				
Operating income (loss)	447	1,285	1,506				

¹ As reported in the Boston Scientific 2018 annual report.

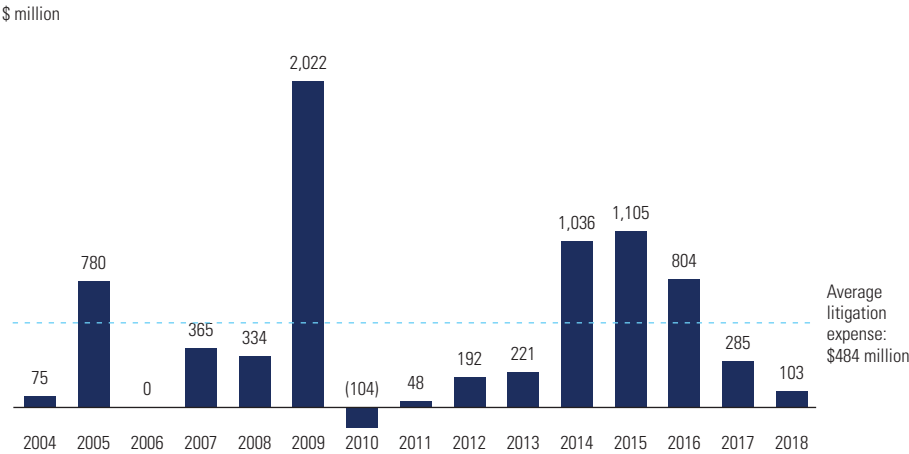
Boston Scientific reports several so-called operating expenses that are in fact nonoperating. Amortization of intangibles (\$599 million in 2018) and intangible-asset impairment charges (\$35 million) are all noncash reductions in the value of intangible assets; they differ only in their timing and regularity. Other nonoperating expenses include contingent consideration benefit (\$21 million), restructuring charges (\$36 million), and litigation-related charges (\$103 million). For valuation purposes, such nonoperating expenses should not be deducted from revenue to determine EBITA.

The right side of Exhibit 21.1 presents the calculation of EBITA for Boston Scientific. Only operating expenses that grow in line with revenue—such as cost of products sold; selling, general, and administrative (SG&A) expense; research and development (R&D) expense; and royalty expense—are included in the calculation of EBITA. Note how the accounting definition of operating income grows dramatically, while the growth in EBITA is much more measured.

As noted earlier, judgment is called for in classifying items as operating or nonoperating. Operating expenses tend to be ongoing and tied to revenue, so a long-term perspective is critical. For instance, treat a plant closure that occurs once in ten years as nonoperating. Conversely, for a retailer with hundreds of stores, treat expenses related to closing stores each year as operating.

For Boston Scientific, we classify royalty payments as operating because royalties are a fundamental part of the medical-devices industry and grow in line with revenue. In contrast, litigation expenses are sporadic and come in waves. Exhibit 21.2 presents litigation expenses for Boston Scientific between 2004 and 2018. While litigation expenses have been declining over the last three years, this trend does not capture the long-term levels. We could treat the litigation expenses as operating, but this would depress ROIC during

EXHIBIT 21.2 **Boston Scientific: Litigation Expenses by Year**



Source: Boston Scientific annual reports.

periods when the expenses were recognized, rather than in the years when the corresponding benefits were reaped. At the same time, litigation expenses are real, so a valuation of Boston Scientific must incorporate them. Although time-consuming, an analysis of the company's current exposure to litigation and an analysis of average litigation expenses across all medical-technology companies could provide valuable insights.

When classification is unclear, measure ROIC with and without the expense. If the expense is lumpy, smooth the expense over the period in which the expense was generated.

Searching the Notes for Hidden One-Time Items

The income statement does not explicitly report every nonoperating expense or one-time charge. These can also be embedded in cost of sales or selling expenses. To find embedded expenses, read the management discussion and analysis section in the company's annual report. The section details the changes in cost of sales and other expenses from year to year and will sometimes report unusual items. In 2011, Boston Scientific reported such an expense:

During the first quarter of 2011, we reversed \$20 million of previously established allowances for doubtful accounts against long-outstanding receivables in Greece. During the first quarter of 2011, the Greek government converted these receivables into bonds, which we were able to monetize, reducing our allowance for doubtful accounts as a credit to selling, general and administrative expenses.

Whether you make an adjustment to NOPAT for such an expense depends on whether the charge is large enough to affect perceptions of performance. If it is not, don't bother. An adjustment could make your analysis overly complex and time-consuming.

Analyzing Each Nonoperating Item for Impact on Future Operations

In Kimberly-Clark's 2018 annual report, the company writes, "The 2018 Global Restructuring Program will reduce our structural cost base by streamlining and simplifying our manufacturing supply chain and overhead organization. The restructuring is expected to generate annual pre-tax cost savings of \$500 to \$550 [million] by the end of 2021." If credible, such projections should be incorporated into your forecast of future cash flow.

More broadly, academic researchers have been examining the predictive component of special items and one-time charges. Early research pointed to the low persistence of special items, indicating that they are in fact transitory and should not be incorporated into forecasts. However, this research examined persistence only on a year-to-year basis. In 2009, researchers extended the window to multiple years and found persistence in special items for companies with strong core profits.¹ In other words, a highly profitable company that reports a series of, say, restructuring charges is likely to continue with similar charges in the future. Persistence was low for companies with little operating profit.

One reason special items may persist year after year for profitable companies is that management may be shifting ongoing operating costs into special items to meet certain earnings targets, as many academic researchers believe they do. This belief also appears common among research analysts, as they decrease their earnings forecasts following the disclosure of a special item.² Although the research showing that special items are used to manage earnings is persuasive, it remains unclear how to relate the research results to an individual company. Judgment is required: pay close attention to companies disclosing special items. If the special items seem likely to recur, especially in a challenging economy, adjust your forecasts accordingly.

A comprehensive list of nonoperating items and one-time charges is impractical, but the following items are the most common: amortization of acquired intangibles; asset write-offs, including write-offs of goodwill and purchased R&D; restructuring charges; litigation charges; and gains and losses on asset sales. Since each of these nonoperating items requires a particular adjustment, we will work through them one by one.

¹ P. M. Fairfield, K. A. Kitching, and V. W. Tang, "Are Special Items Informative about Future Profit Margins?" *Review of Accounting Studies* 14, nos. 2–3 (2009): 204–236.

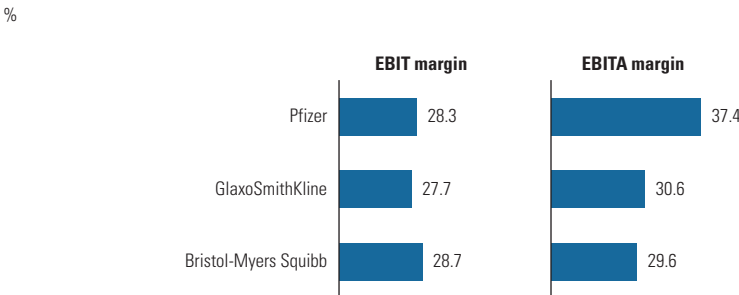
² N. Li, H. Su, W. Dong, and K. Zhu, "The Effect of Non-recurring Items on Analysts' Earnings Forecasts," *China Journal of Accounting Research* 11, no. 1 (2018): 21–31.

Amortization of Acquired Intangibles Although accounting standards require amortization of acquired intangibles, in most circumstances you should *not* deduct amortization from operating profit to determine NOPAT. As an alternative to expensing amortization, use EBITA (not EBIT) to determine operating profits. Since amortization is excluded from operating profit, remember to include the cumulative excluded amortization in your total for intangible assets on the balance sheet. A corresponding entry should be made to equity (titled “cumulative amortization”) to balance total funds invested.

Why not amortize intangibles, particularly since we include depreciation in our calculation of ROIC? The idea of recognizing an intangible asset and then amortizing its use over a useful life is a good one. Yet current accounting standards do not allow companies to take this approach consistently across all intangibles. Today, only *acquired* intangibles are capitalized and amortized, while *internally generated* intangible assets, such as brand and distribution networks, are expensed when they are created. Thus, the EBIT of a company that acquires an intangible asset and then replenishes the asset through internal investment will be penalized twice on its financial statements, once through SG&A expenses and again through amortization. In fact, expensing the creation of new intangible assets while amortizing old intangibles would be tantamount to including both capital expenditures and depreciation on the income statement, a clearly undesirable characteristic. For valuation purposes, avoid mixing amortization and expensing by maintaining goodwill and acquired intangibles at their original values. To do this, compute operating profit before amortization, and add cumulative amortization to the current value of goodwill and intangible assets.

Exhibit 21.3 demonstrates the effect of amortizing acquired intangibles on margins for three companies in the pharmaceuticals industry. Based on EBIT margin, it appears as if the three companies have nearly identical performance. The amortization of acquired intangibles, however, is distorting our perspective. Pfizer has been extremely active in acquiring companies and

EXHIBIT 21.3 **EBIT and EBITA Margins in the Pharmaceuticals Industry, 2018**



Source: Annual reports.

products, including the 2016 purchases of Medivation and Anacor. Stripping out amortization from these and other acquisitions reveals that Pfizer outperformed these peers by roughly seven percentage points.

One situation in which it is appropriate to deduct amortization is when intangibles can be capitalized (versus expensed) consistently. Consider a company that has no sales force and instead purchases customer contacts from a third party. Since sales outlays are never expensed via SG&A, they should be amortized to arrive at a meaningful measure of operating profitability. Otherwise, the income statement would not accurately reflect the cost of selling. Another example is the purchase of frequency rights by telecom companies. Since these assets can be capitalized and amortized without exception, treat them no differently than fixed equipment and depreciation.

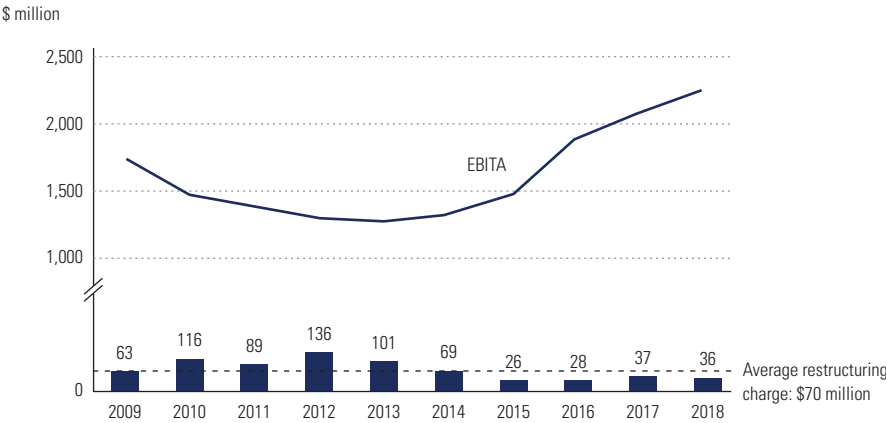
Asset Write-Offs If the value of an asset falls below its book value, accounting standards dictate that the asset should be written down (sometimes entirely) to its fair value. Although write-downs and write-offs give lenders insight into the diminished value of their collateral, the resulting balance sheet value understates the historical investment made by shareholders. Thus, ROIC can artificially rise following a write-down. To counteract this effect, treat asset write-downs and write-offs as nonoperating, and *add* cumulative write-downs to invested capital. To balance total funds invested, create a corresponding equity equivalent.

Two categories of asset write-offs are common:

1. *Asset write-offs.* In general, treat an asset write-off as nonoperating. In the rare cases when they occur systematically, treat them as operating. Add back write-downs to the asset, except when you are estimating capital turnover to project future capital needs. In this case, compute the ratio in a manner that best reflects future capital needs.
2. *Goodwill and intangibles impairments.* Treat goodwill and other intangibles impairments as nonoperating and add back cumulative impairments to goodwill on the balance sheet. Since the purpose of computing ROIC with goodwill is to measure historical performance *including* all past acquisition premiums, goodwill should remain at its original level.

Restructuring Charges As business changes, companies must adapt. Major changes often require plant closures, employee layoffs, inventory write-downs, asset write-offs, and other restructuring charges. If a restructuring charge is unlikely to recur, treat the charge as nonoperating. If, however, a pattern of ongoing restructuring charges emerges, further analysis is required. Exhibit 21.4 presents the restructuring charges for Boston Scientific between 2009 and 2018. During this period, Boston Scientific's restructuring charges averaged

EXHIBIT 21.4 **Boston Scientific: EBITA and Restructuring Charges**



Source: Boston Scientific annual reports.

\$70 million per year, or 0.9 percent of revenues. These expenses are reported separately from cost of sales and SG&A.

Given their persistence, Boston Scientific’s restructuring charges should be analyzed to determine what portion of them represents cash (such as severance payments), whether any cash restructuring charges are likely to continue, and for how long. To this end, a careful reading of the company’s notes reveals the following:

In November 2018, the Board of Directors approved, and we committed to, a new global restructuring program (the 2019 Restructuring Plan). The 2019 Restructuring Plan is expected to result in total pre-tax charges of approximately \$200 million to \$300 million and reduce gross annual pre-tax operating expenses by approximately \$100 million to \$150 million by the end of 2022 as program benefits are realized.

Many restructuring charges are recorded before any cash is spent. If this is the case, a corresponding reserve will be recorded in the liabilities section of the balance sheet. In the next main section, we consider treatment of various reserves, including those related to restructuring charges.

Litigation Charges When there is likely to be a legal judgment against a company, the company will recognize a litigation charge. If the litigation charge recurs frequently and grows with revenue, treat the charge as operating. For instance, hospital systems frequently defend themselves against malpractice lawsuits. Since these lawsuits are a cost of doing business, the litigation costs should be treated as operating costs for valuation and projected

forward. However, if a litigation cost is truly a one-time expense, treat it as nonoperating, and value any claims against the company separately from core operations.

Gains and Losses on Asset Sales When an asset's sale price differs from its book value, the company will recognize a gain or loss. Since current gains and losses are backward-looking (value has been created or destroyed in the past), treat them as nonoperating. Additionally, double-check to make sure projected free cash flow does not incorporate the asset recently sold. For instance, make sure future depreciation reflects only the remaining assets.

Although gains and losses should not be included in operating profit, past asset sales may provide insight about the level of cash to be generated by future asset sales. Again, be careful to value future asset sales (and their corresponding gains and losses) only when the assets are not incorporated in free cash flow. Otherwise, the resulting double-counting will overstate the company's value.

PROVISIONS AND THEIR CORRESPONDING RESERVES

Provisions are noncash expenses that reflect future costs or expected losses. Companies take provisions by reducing current income and setting up a corresponding reserve as a liability (or deducting the amount from the relevant asset).

For the purpose of analyzing and valuing a company, we categorize provisions into one of four types: ongoing operating provisions, long-term operating provisions, nonoperating provisions like restructuring provisions, or provisions created for the purpose of smoothing income (transferring income from one period to another). Based on the characteristics of each provision, adjust the financial statements to reflect the company's true operating performance. For example, ongoing operating provisions are treated the same way as any other operating expense, whereas restructuring provisions are converted from an accrual to a cash basis and treated as nonoperating. Exhibit 21.5 summarizes the four types of provisions and how to treat them in NOPAT, invested capital, and valuation.

We believe our classification system of reserves leads to better analysis, but the way you adjust the financial statements should not affect the company's valuation. The valuation depends solely on how and when cash flows through the business, not on accrual-based accounting.

Adjustments for the Provisions

In Exhibit 21.6, we present the abbreviated financial statements for a hypothetical company that recognizes four provisions: an environmental provision

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EXHIBIT 21.5 **Treatment of Provisions and Reserves**

Classification	Examples	Treatment in NOPAT ¹	Treatment in invested capital	Treatment in valuation
Ongoing operating provisions	Product returns and warranties	Deduct provisions from revenue to determine NOPAT.	Deduct reserve from operating assets to determine invested capital.	Provision is part of free cash flow.
Long-term operating provisions	Plant decommissioning costs and unfunded retirement plans	Deduct operating portion from revenue to determine NOPAT, and treat interest portion as nonoperating.	Treat reserve as a debt equivalent.	Deduct reserve's present value from the value of operations.
Nonoperating provisions	Restructuring charges, such as expected severance due to layoffs	Convert accrual provision into cash provision, and treat as nonoperating.	Treat reserve as a debt equivalent.	Deduct reserve's present value from the value of operations.
Income-smoothing provisions	Provisions for the sole purpose of income smoothing	Eliminate provision by converting accrual provision into cash provision.	Treat reserve as an equity equivalent.	Since income-smoothing provisions are noncash, there is no effect.

¹ Net operating profit after taxes.

EXHIBIT 21.6 **Provisions and Reserves in the Financial Statements**

\$ million

	Today	Year 1	Year 2	Year 3	Year 4
Income statement					
Revenue		1,000.0	1,200.0	1,400.0	1,600.0
Operating costs		(750.0)	(900.0)	(1,190.0)	(1,200.0)
Decommissioning asset, depreciation ¹		(7.7)	(7.7)	(7.7)	—
Decommissioning reserve, accretion ²		(15.0)	(16.5)	(18.2)	—
Provision for product defects ²		(100.0)	(120.0)	(140.0)	(160.0)
Income-smoothing provision ²		(40.0)	(40.0)	80.0	—
Operating profit, as reported		87.3	115.8	124.1	240.0
Provision for restructuring		—	(30.0)	—	—
Net income		87.3	85.8	124.1	240.0
Balance sheet					
Decommissioning asset, gross	77.1	77.1	77.1	77.1	—
Accumulated depreciation	(54.0)	(61.7)	(69.4)	(77.1)	—
Decommissioning asset, net	23.1	15.4	7.7	—	—
Other operating assets	700.0	840.0	980.0	1,120.0	—
Total assets	723.1	855.4	987.7	1,120.0	—
Reserve for decommissioning	150.3	165.3	181.8	—	—
Reserve for product defects	100.0	120.0	140.0	160.0	—
Reserve for restructuring	—	—	30.0	—	—
Reserve for income smoothing	—	40.0	80.0	—	—
Equity	472.9	530.1	555.9	960.0	—
Total liabilities and equity	723.1	855.4	987.7	1,120.0	—

¹ Typically embedded in depreciation and amortization.

² Typically embedded in operating costs, such as cost of sales.

EXHIBIT 21.7 ROIC with Provisions and Reserves

\$ million

	Today	Year 1	Year 2	Year 3	Year 4
NOPAT					
Operating profit, as reported		87.3	115.8	124.1	240.0
Decommissioning reserve, accretion		15.0	16.5	18.2	—
Increase (decrease) in income-smoothing reserve		40.0	40.0	(80.0)	—
NOPAT		142.3	172.3	62.3	240.0
Reconciliation to net income					
Net income		87.3	85.8	124.1	240.0
Decommissioning reserve, accretion		15.0	16.5	18.2	—
Increase (decrease) in income-smoothing reserve		40.0	40.0	(80.0)	—
Provision for restructuring		—	30.0	—	—
NOPAT		142.3	172.3	62.3	240.0
Invested capital					
Plant decommissioning, net	23.1	15.4	7.7	—	—
Other operating assets	700.0	840.0	980.0	1,120.0	—
Reserve for product defects	(100.0)	(120.0)	(140.0)	(160.0)	—
Invested capital	623.1	735.4	847.7	960.0	—
Reconciliation of invested capital					
Reserve for plant decommissioning	150.3	165.3	181.8	—	—
Reserve for restructuring	—	—	30.0	—	—
Reserve for income smoothing	—	40.0	80.0	—	—
Equity	472.9	530.1	555.9	960.0	—
Invested capital	623.1	735.4	847.7	960.0	—
ROIC on beginning-of-year capital, %					
		22.8	23.4	7.3	25.0

for decommissioning the company's plant, an operating provision for future product defects, a provision for smoothing income, and a restructuring provision for future severance payments. In this example, we reorganize forecast statements, rather than historical statements, to demonstrate how each type of provision would be treated from a valuation perspective. (Historical statements should be adjusted in the same way as forecast statements.) For simplicity, we assume the company pays no taxes and has no debt.

The process for adjusting the financial statements depends on the type of provision. Exhibit 21.7 shows how to reorganize the income statement and balance sheet for each provision for our hypothetical company. In the following discussion of this example, all numbers in parentheses refer to the year 1 reorganized financial statements.

Provisions Related to Ongoing Operations When a company warrants a product, expects that some products will be returned, or self-insures a service, it must create a corresponding liability when that product or service is sold. If

the reserve is related to the ongoing operations, the reserve should be treated the same way as other non-interest-bearing liabilities (e.g., accounts payable and wages payable). Specifically, the provision should be deducted from revenues to determine EBITA. The corresponding reserve (\$100 million) should be netted against operating assets (\$723.1 million) to measure invested capital (\$623.1 million). Since the provision and reserve are treated as operating items, they appear as part of free cash flow and should not be valued separately.

Long-Term Operating Provisions Sometimes, when a company decommissions a plant, it must pay for cleanup and other costs. Assume our hypothetical company owns a plant that will operate for ten years and requires \$200 million in decommissioning costs. Rather than expense the cash outflow in a lump sum at the time of decommissioning, a company will instead record the present value of the cost as both an asset and a liability at the time of investment.³ In this case, the ten-year present value of \$200 million at 10 percent equals \$77.1 million.⁴ It's as if the company borrowed \$77.1 million and holds the money in restricted cash to fund the future decommissioning outlay.

Once the decommissioning asset and reserve are recognized, the decommissioning asset is depreciated (similar to the way restricted cash is paid into an outside fund set aside for cleanup), and the reserve is grown (as if the debt accumulates unpaid interest charges). As a result, the decommissioning cost is recognized over the life of the asset, instead of a lump sum at closing.

If the decommissioning costs are substantial, as with a nuclear power plant or a mine, the costs will be presented in the company's footnotes. We show a sample note in Exhibit 21.8. In Panel A of Exhibit 21.8, the decommissioning asset declines by \$7.7 million each year. This expense is computed using straight-line depreciation on the original decommissioning asset. In Panel B, the decommissioning reserve grows each year by an ever-increasing amount, computed at 10 percent of the prior year's ending reserve. This expense, which mimics interest, is known as accretion. In year 1, the current-year reserve of \$150.3 million grows by \$15.0 million in accretion. The income statement presented in Exhibit 21.6 reports both depreciation and accretion as operating items, often embedded within depreciation and operating costs, respectively.

To estimate NOPAT, invested capital, ROIC, and FCF, apply the guiding principles presented in Chapter 11. When reorganizing the income statement,

³ In the United States, asset retirement obligations (AROs) are governed by SFAS 143. Entities covered by IFRS use IAS 37, where the AROs are called "provisions."

⁴ In Exhibit 21.6, the current year represents the seventh year of the plant's expected ten-year life. Consequently, the decommissioning asset and the decommissioning reserve no longer equal their initial value of \$77.1 million. Instead, the decommissioning asset has been depreciated seven years to \$23.1 million, using straight-line depreciation. Conversely, the decommission reserve grows annually at the discount rate. As a result, the current year reserve equals $\$77.1 \text{ million} \times (1.10)^7$, which equals \$150.3 million.

EXHIBIT 21.8 Provisions and Reserves in the Notes

\$ million

	Today	Year 1	Year 2	Year 3
Panel A: Change in the asset account				
Decommissioning asset, starting	30.8	23.1	15.4	7.7
Depreciation	(7.7)	(7.7)	(7.7)	(7.7)
Decommissioning asset, ending	23.1	15.4	7.7	—
Panel B: Change in the liability account				
Decommissioning reserve, starting	136.6	150.3	165.3	181.8
Accretion expense at 10%	13.7	15.0	16.5	18.2
Payout	—	—	—	(200.0)
Decommissioning reserve, ending	150.3	165.3	181.8	—
Panel C: Income statement				
Decommissioning asset, depreciation	7.7	7.7	7.7	—
Decommissioning reserve, accretion	15.0	16.5	18.2	—
Decommissioning expense	22.7	24.2	25.9	—

treat depreciation as an operating item. Conversely, since accretion mimics interest, do not include accretion in NOPAT; instead, include it in the reconciliation to net income, next to interest expense. NOPAT and the reconciliation to net income are computed in the top portion of Exhibit 21.7. To reorganize the balance sheet, classify the decommissioning asset (which is comparable to restricted cash) as part of invested capital and the reserve as a debt equivalent. Invested capital and its reconciliation are presented in the bottom half of Exhibit 21.7. To compute free cash flow, start with NOPAT, add back depreciation of the decommissioning asset (since it is noncash), and subtract investments in invested capital. Free cash flow is presented in Exhibit 21.9.

When you treat the plant closure reserve as a debt equivalent, the interest expense and reserve drawdown will not flow through free cash flow. Therefore, subtract the current reported reserve (\$150.3 million as of today) from the value of operations (\$858.9 million) to determine equity value. The value of operations, which includes this and other deductions, is converted into equity value in Exhibit 21.10.

One-Time Restructuring Provisions When management decides to restructure a company, it will often recognize certain future expenses (e.g., severance payments) immediately. We recommend treating one-time provisions as non-operating and treating the corresponding reserve as a debt equivalent. In year 2, our hypothetical company declared a \$30 million restructuring provision, which will be paid in year 3 (see Exhibit 21.6). Since the restructuring is non-operating, it is not deducted from revenues to determine NOPAT. Rather, it is included in the reconciliation to net income (see Exhibit 21.7). Because we plan to value the provision on a cash basis, the noncash reserve is treated as a debt

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EXHIBIT 21.9 **Free Cash Flow with Provisions and Reserves**

\$ million					
	Year 1	Year 2	Year 3	Year 4	
NOPAT	142.3	172.3	62.3	240.0	
Depreciation	7.7	7.7	7.7	—	
Gross cash flow	150.0	180.0	70.0	240.0	
Investment in invested capital	(120.0)	(120.0)	(120.0)	960.0	
Free cash flow	30.0	60.0	(50.0)	1,200.0	Present value at 10% = 858.9
Reconciliation of free cash flow					
Provision for restructuring	—	30.0	—	—	
(Increase) decrease in restructuring reserve	—	(30.0)	30.0	—	
Cash-based restructuring provision	—	—	30.0	—	Present value at 10% = 22.5
Decommissioning reserve, accretion	15.0	16.5	18.2	—	
(Increase) decrease in decommissioning reserve	(15.0)	(16.5)	181.8	—	
Dividends	40.0	70.0	—	1,120.0	
Equity repurchases (issues)	(10.0)	(10.0)	(280.0)	80.0	
Free cash flow	30.0	60.0	(50.0)	1,200.0	

equivalent and therefore is not netted against operating assets to determine invested capital.

Since nonoperating income and expenses do not flow through free cash flow, the restructuring expense must be valued separately on a cash basis. To convert accrual-based restructuring expenses to cash, start with the restructuring expense, and subtract the increase in the restructuring reserve. This leads to a cash-based restructuring provision of \$0 in year 2 and \$30 million in year 3 (free cash flow and its reconciliation are presented in Exhibit 21.9). The present value of the nonoperating cash flow stream equals \$22.5 million, which must be deducted from the value of operations to determine equity value, as shown in Exhibit 21.10.

Income-Smoothing Provisions Except for limited circumstances, provisions to smooth earnings are not allowed under International Financial Reporting Standards or U.S. Generally Accepted Accounting Principles (GAAP). To

EXHIBIT 21.10 **Enterprise DCF with Provisions and Reserves**

\$ million		
Valuation		Methodology
Value of operations	858.9	Summation of discounted cash flow
Value of restructuring provision	(22.5)	Present value at 10% (debt equivalent)
Reserve for plant decommissioning	(150.3)	Reported on balance sheet (debt equivalent)
Equity value	686.1	

prevent earnings manipulation or even the perception of it, many companies use a third party to estimate key provisions. In some situations, companies can use provisions to smooth earnings. For instance, defense contractors will use income smoothing when they believe a long-term contract's value has changed.

In Exhibit 21.6, our hypothetical company was able to show a smooth growth in reported EBITA and net income by using a smoothing provision. We choose a straightforward title for the account, "Income-smoothing provision," but actual companies typically use subtler wording, such as "Other provisions." For our hypothetical company, a provision was recorded in years 1 and 2 and was reversed in year 3. By using an income-smoothing provision, the company hid its year 3 decline in operating performance (operating costs rose from 75 percent to 85 percent of sales).

To evaluate the company's performance properly, eliminate any income-smoothing provisions. Do this by adding the income-smoothing provision back to reported EBITA (essentially undoing the income-smoothing provision). In this way, we are converting the provision to cash, rather than accounting for it as an accrual, and subsequently need to treat the reserve as an equity equivalent (using a process identical to the one for deferred taxes). Since income-smoothing provisions are entirely noncash, they don't affect free cash flow or valuation.

Provisions and Taxes

In most situations, provisions are tax deductible only when cash is disbursed, not when the provision is reported. Thus, most provisions will give rise to deferred-tax assets. For example, a \$30 million noncash restructuring charge would lead to a \$30 million restructuring reserve. If the restructuring charge is tax deductible on the GAAP income statement, retained earnings would drop by only \$21 million (assuming a 30 percent tax rate). Since the increase in the restructuring reserve does not match the drop in retained earnings, the balance sheet will not balance. To eliminate the difference, a deferred-tax asset is recognized for \$9 million.

For operating-related provisions, we recommend using cash, rather than accrual taxes. For nonoperating provisions, estimate the tax impact of the corresponding provision. Do not use book values, as they reflect past accounting and not necessarily future cash flow. For an in-depth discussion of deferred taxes, see Chapter 20.

CLOSING THOUGHTS

The accounting definition of nonoperating expense is narrow and limited to interest expense and a few other items. Therefore, the accounting definition of operating profits will inappropriately include many one-time and other

nonoperating items. Always start your financial analysis by separating operating from nonoperating items on the income statement. This will create a better picture of the company's performance and its potential for generating future cash flow. In some cases, the proper classification of a particular expense will be unclear. But don't let this distract you from the task at hand. A proper valuation will not depend on how the item is treated, as long as you include it somewhere and treat it consistently.

REVIEW QUESTIONS

1. What are the three steps to assess the impact of nonoperating expenses and one-time charges on cash flow projections?
2. ValueCo generates \$10 million in after-tax operating profit on \$100 million in operating assets. The company has \$20 million in accounts payable, \$15 million in product warranty reserves, \$5 million in severance reserves, \$30 million in long-term debt, and \$30 million in equity. What is ValueCo's return on invested capital?
3. In year 0, SmoothCo has \$50 million in cash and \$50 million in inventory, financed by \$100 million in equity. In year 1, the company records \$100 million in revenue, \$80 million in operating costs, and \$10 million in litigation provisions for a case yet to be resolved. Based on the preceding data, build an income statement and balance sheet for year 1. Assume inventory remains constant, no dividends are paid, and no taxes. What is the return on equity (ROE) in year 1? In year 2, the company records \$100 million in revenue and \$90 million in operating costs. The case started in year 1 is resolved for \$5 million in cash. Because management overestimated the amount of litigation charges, SmoothCo takes a \$5 million gain in year 2. What is ROE in year 2? How is ROE distorted by the litigation expense?
4. Companies in highly competitive industries often see a number of consecutive restructuring charges. In these cases, should restructuring be treated as operating or nonoperating? From a valuation perspective, what are the important issues that should be considered?
5. What are some of the more common nonoperating items and one-time charges that should be excluded from operating expenses?

